

# ***Ancylostoma duodenale***

# ***Necator americanus***

## **(Hookworms)**

*GIT module- Microbiology- LG5*

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- *Ancylostoma duodenale* (old world hookworm), *Necator americanus* (new world hookworm) are the major etiology of human hookworm infection.
- Disease is known as ancylostomiasis, necatoriasis or hookworm infection.
- The adult worms live in the lumen of small intestine of the host (commonly, jejunum). They remain attached to the intestinal wall by their buccal cavity.
- *N. americanus* is the predominant hookworm worldwide, with the highest rates in Sub-Saharan Africa, tropical regions of the Americas, China, and Southeast Asia. *A. duodenale* is more endemic in parts of China, India, North Africa, Sub-Saharan Africa, and a few regions of the Americas.

- Hookworms infect close to 500 million people in the developing nations of the tropics.
- Hookworm infection is the most common causes of iron- deficiency anemia in endemic countries.
- Human faeces is the main source of infection.
- Man is only reservoir for both spp.
- *Ancylostoma ceylanicum*, is found mainly in cats, but human cases have been reported in Malaysia and elsewhere in Asia.

## **Routes of infection**

- 1- Cutaneous route : penetration of the skin (particularly skin of feet, hands, buttocks) by filariform larvae (most common way).
- 2- Oral route: ingestion of the filariform larvae with contaminated soil (less frequent).
- 3- Trans-mammary transmission (rarely).
- 4- Trans-placental transmission (very rare).

# Morphology

## Adult worms

The adult worms are cylindrical, greyish white and slightly curved. The anterior end is bent slightly in the same direction of the body curve giving hook shape. The mouth or buccal capsule is lined with a 6 chitinous material (teeth or cutting plates).

The male (8 mm) is relatively smaller than the female (12 mm the posterior end is straight). The posterior end of the male is equipped with a bell-like expansion known as copulatory bursa which contain 13 finger-like rays in 3 lobes. The adults live for a period of 2-3 years. The female produce 10000-25000 eggs per day during this period.

The buccal capsule of *A. duodenale* is large and oval, contains 2 pairs of claw-like teeth on the ventral surface and 1 pair of knob like teeth on the dorsal surface.

In *N. americanus* the buccal capsule is small and round contains 2 pair of cutting plates on the ventral surface and 1 pair of plates on the dorsal surface.

## **Eggs**

Measure 60\*40  $\mu\text{m}$ , surrounded by a thin, and transparent shell. The eggs usually contain 7-8 blastomeres. Clear space is always present between the segmented ovum and the egg shell.

## **Rhabditiform larvae**

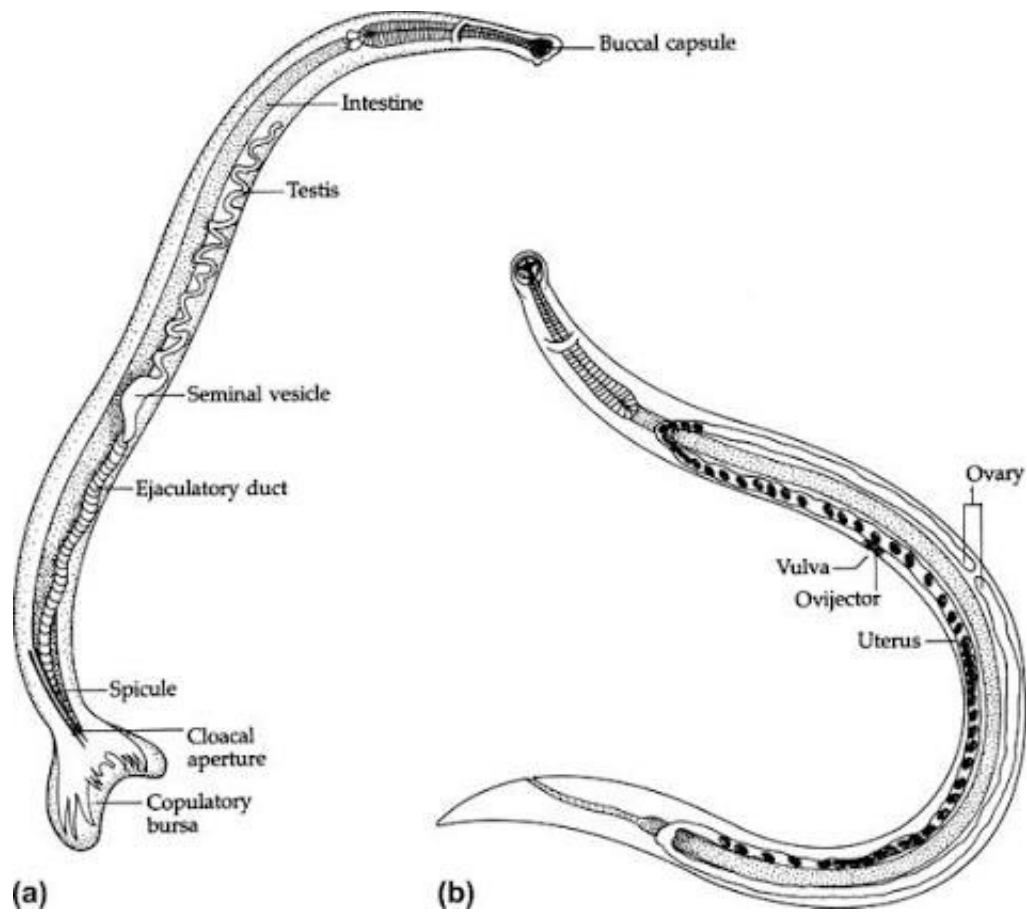
Is the first stage larva which immediately hatch out of the eggs. They are feeding, actively motile and measure 200-300  $\mu\text{m}$  in length \* 16  $\mu\text{m}$  in width. Each larva has a short mouth, bulbed esophagus, and conspicuous genital primordium.

## **Filariform larvae**

They are non-feeding, cylindrical, measure 500-600  $\mu\text{m}$  in length with a much shorter esophagus.

# Morphology



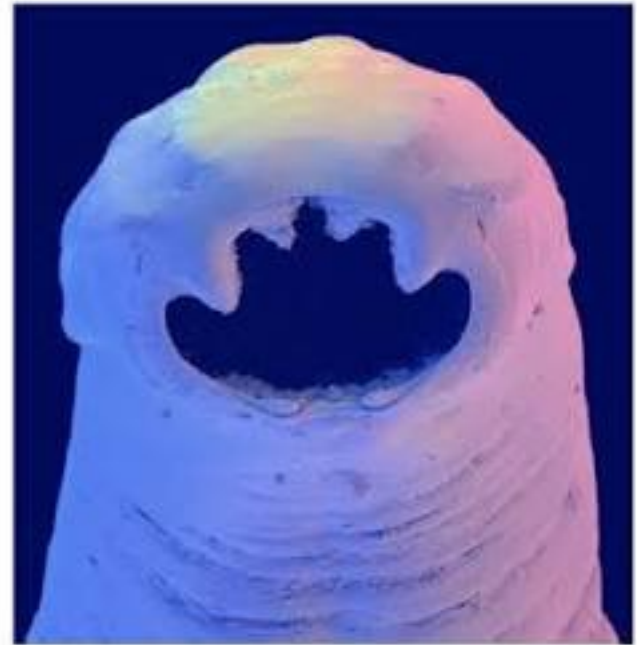


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*Ancylostoma duodenale*  
a: Male; b: female; c: egg



*Necator americanus*

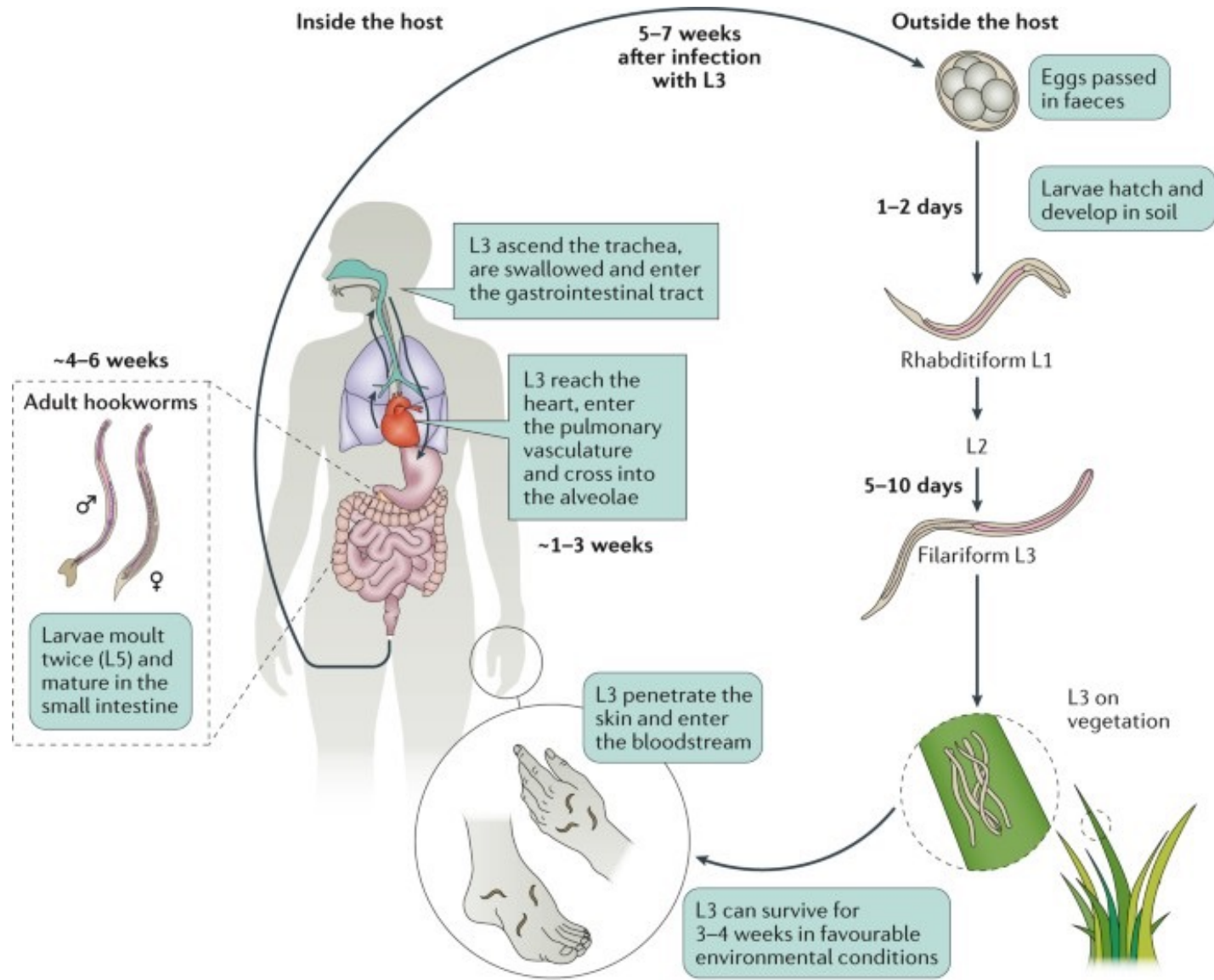


*Ancylostoma duodenale*

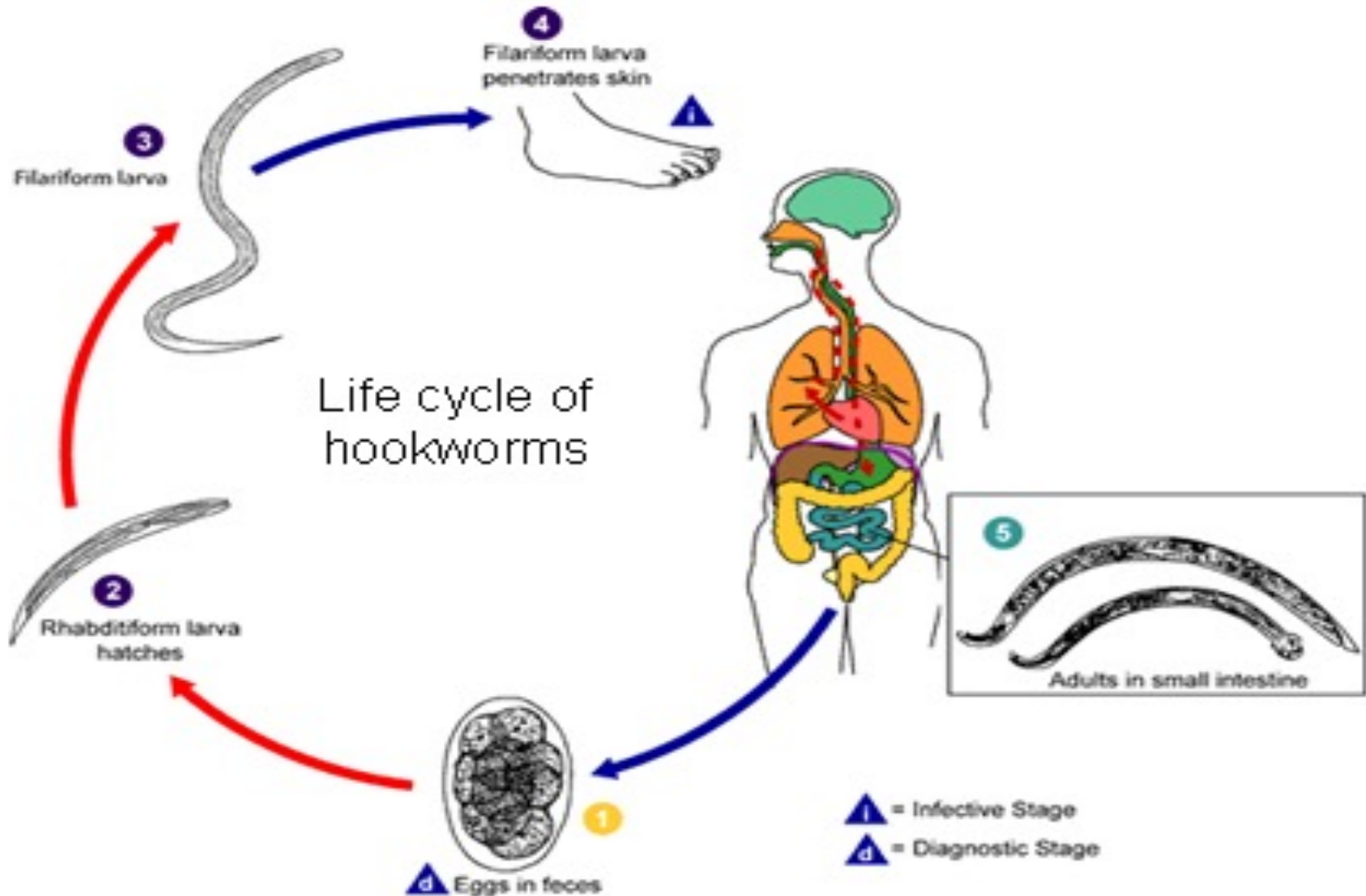


# Life cycle

Hookworms are soil-transmitted helminths. Life cycle is completed in a single host. Man is the only host. Man acquires infection following penetration of the skin by filariform larvae. These larvae shed off their sheaths and by mechanical and lytic actions enter through the epidermis into the dermis and subcutaneous tissues. From these deeper tissues they enter the circulation and carried to the right heart and from there to lungs. The larvae break out from the capillaries and migrate up to the pharynx, swallowed and come back to the small intestine. When the larvae pass through the esophagus they molt once and develop into L4 which equipped with buccal cavity help the larvae to adhere to the gut wall. In the small intestine, these larvae molt twice more and develop to adult worms. Fertilization occurs, the fertilized females begin to lay eggs in about 4-7 weeks after infection (penetration of the skin by the larvae). The eggs passed with the faeces. In the soil and under suitable conditions, they develop and hatch to rhabditiform larvae (within 24-48 hr). The released larvae molt twice and develop into filariform larvae (the infective stage) in about 5-10 days which can survive in the soil for about 3-4 weeks.



Life Cycle of hookworms



Life Cycle of hookworms

## Pathogenesis

Infection with *A. duodenale* is more serious than that caused by *N. americanus*. The pathological lesions are mainly caused by the adults and less frequently caused by the larvae.

The major pathological lesions are due to the attachment of the adult worms to the intestinal wall by their buccal cavity, provided by strong chitinous teeth which cut the blood vessels and feed on blood and tissue fluid causing blood loss.

Each worm cause a daily blood loss of 0.15-0.26 ml (*A. duodenale*) and 0.03 ml (*N. americanus*). A heavy *A. duodenale* infection can produce total daily losses of 10 to 20 mL of blood.

Blood loss caused by:

- 1- Direct absorption of blood.
- 2- Bleeding of the affected sites.

Hookworms secrete some anticoagulant factors which increase the clotting time and prolong the blood seepage.

Excessive blood loss caused by a heavy and prolonged infection lead to iron deficiency anemia which can frequently become serious and even fatal in persons with defects in normal iron intake. Loss of protein, results in hypoproteinemia and oedema.

Hookworms produce a serine protease inhibitor that interferes with pancreatic enzyme-based digestion, may also interfere with breakdown of ingested foods, leading to malabsorption and consequent retardation of growth and development.

The resulting iron deficiency and malnutrition during pregnancy adversely affect intrauterine growth, birth weight, and even maternal survival.

## **Pathogenicity of the larvae**

The filariform larvae at the site of the penetration produce local reaction called ground itch which is an allergic manifestation.

Migration of a large number of larvae through the lung produces minute haemorrhage, leukocyte infiltration, eosinophilia and leukocytosis.

## **Clinical manifestation**

In endemic areas about 20-40 % of the cases are asymptomatic. Clinical manifestation in symptomatic cases are variable depending on the stages of infection:

### **A- Skin manifestation**

Ground itch is the most important and earliest symptom developed in about 7-10 days at the site of larva penetration. This condition characteristically seen in feet, particularly between toes, characterized by intense itching, oedema, erythema and rash. Secondary bacterial infection aggravate the condition. Ground itch is frequently seen in persons with previous exposure to the parasite.

## **B- Respiratory manifestation**

Migration of a large number of larvae through the lung produces: low-grad fever, mild cough, dyspnea, pharyngitis and occasionally hemoptysis. Pneumonia and bronchitis are seen in persons with sever infection.

## **C- Intestinal manifestation**

Intestinal stage of the disease manifests as low grade fever, nausea, vomiting, diarrhea, flatulence and abdominal discomfort.

Iron deficiency anemia and hypoalbuminemia are the major clinical manifestation.

## **Immunity to hookworm infection**

Repeated exposure to hookworm infection does not necessarily result in protective immunity.

Although a humoral antibody response to numerous hookworm antigens can be demonstrated in naturally infected individuals, the presence of specific antibodies often does not correlate with resistance to infection.

Some immunoglobulins may even serve as a marker for hookworm infection.

The absence of effector immunity may explain why the intensity of hookworm infection often increases with age in endemic regions.

# **Hookworms molecules involved in immuno-evasion**

## **1- Neutrophil inhibitory factor**

Cysteine rich glycoproteins block the adhesion of activated human neutrophils to vascular endothelial cells and also inhibits the release of  $\text{H}_2\text{O}_2$ .

## **2- Proteases**

Hookworms produce some proteases such as elastin-cleaving metalloproteases, which cleave eotaxin and inhibit eotaxin mediated recruitment of eosinophil to sites of inflammation.

Excretory/ secretory products of *N. americanus* contains proteases capable of cleaving IgA.

### **3- C-type lectins**

Are a family of glycoproteins that act as receptors for glycoprotein ligands. Found in the serum and on the surface of endothelial cells and immune effective cells (APC, B lymphocytes, T lymphocytes). These lectins play a major role in inflammatory and immune response. The parasite derived c-type lectins might be involved in immuno-evasion by competing with host lectins for specific ligands (P-selectin, CD 23)

## **4- Antioxidants**

Nematodes secrete antioxidant enzymes that might protect exposed cuticular surface by neutralizing host free radicals generated by oxidative burst of leukocytes. Superoxide dismutase and glutathione-S transferase activities have been detected in the extracts of hookworms.

# **Diagnosis**

Hookworm infection is difficult to diagnose clinically, but eosinophilia and hypochromic microcytic anemia may be suggestive of the infection in the endemic areas.

## **Laboratory diagnosis**

1- Direct microscopic examination of stool sample for characteristic eggs. This method is adequate to detect moderate and severe infections. In light infection concentration methods are useful. Kato-Katz technique can be used to estimate the intensity of the infection by counting the number of eggs in known amount of the stool sample.

2- Stool culture can be done for detection of larvae. Harada-Mori technique is used for stool culture.

## **Treatment**

Mebendazole is the drug of choice for treatment of hookworm infection. Its effective against both adults and larvae. 100 mg twice daily, given orally for 3 days.

- Pyrantel pamoate
- Thiabendazole
- Levamisole

## **Prevention and control**

- 1- Sanitary disposal of human faeces.
- 2- Treatment of infected individuals.
- 3- Use of latrines.
- 4- Avoidance of walking bare foot in endemic areas.
- 5- Health education.

## ***Ancylostoma Caninum* and *A. braziliense***

are hookworms of dogs and cats. Man is not suitable host for these parasite. In humans these parasites are unable to develop into adults, instead the filariform larvae wander in the superficial layer of the skin of feet, legs, thigh and buttock causing a condition called creeping eruption. These larvae usually die in the skin but occasionally migrate to the lung causing pulmonary manifestation.

# References

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**Thank you**